2025 IEEE 12th International Conference on Cyber Security and Cloud Computing (CSCloud) | 979-8-3315-8781-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/CSCLOUD66326.2025.00041

MGA-KGF: A Multi-Granularity Adaptive Knowledge Graph Fusion Framework for Heterogeneous Multi-Source Smart Grid Data

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***Abstract*—Smart grids generate heterogeneous data streams that span real-time operation and long-horizon planning. However, Transaction Processing (TP) data and Analytical Processing (AP) data differ markedly in timeliness, structure, and semantics, creating persistent barriers to end-to-end intelligence. We propose MGA-KGF, a multi-granularity, layered framework for knowledge-graph-centric fusion of heterogeneous multi-source smart-grid data. MGA-KGF (i) performs unified multi-source access and quality assurance, (ii) constructs fine-grained, medium-grained, and coarse-grained knowledge units with explicit spatio-temporal alignment, and (iii) realizes bidirectional TP/AP fusion via an event-driven TransE–GAT pipeline with temporal and statistical attention. A load-aware storage and indexing scheme dynamically balances query latency and compression by migrating hot and cold knowledge. Deployed across core scenarios—equipment lifecycle management, rapid fault localization and traceability, regional load forecasting and resilience assessment, and O&M risk evaluation—MGA-KGF improves timeliness, semantic consistency, and cross-granularity reasoning quality, while remaining modular and extensible. The framework offers a practical path toward knowledge-reliable, real-time-to-planning integration in modern power systems and lays the groundwork for future incorporation of large language models and renewable-rich operating contexts.**

***Index Terms*—Knowledge graph fusion, multi-granularity modeling, smart grid, heterogeneous multi-source data, TP/AP data integration**

I. Introduction

With the acceleration of the global energy transition [1, 2] and the continuous advancement of smart-grid technologies, power systems [3, 4] have evolved from traditional single-energy transmission networks into complex systems characterized by multi-source synergy, real-time interactivity and intelligent decision-making. In this context, data generated during grid operation are increasing exponentially. Data spanning the entire electricity-supply chain—generation, transmission, distribution and consumption—now constitutes a core strategic asset for ensuring the security and stability of power systems and for optimizing resource allocation [5].

The operation and management of modern smart grids rely on two distinct classes of data resources: transactional processing (TP) data and analytical processing (AP) data. These classes differ substantially in their characteristics and application scenarios, and their efficient integration is a prerequisite for intelligent grid management.

TP data arise from the real-time operation and monitoring of power systems. They are characterized by high concurrency, stringent timeliness and small packet size; processing latencies must often be reduced to milliseconds or even microseconds. Typical TP data include real-time voltage and current measurements acquired by supervisory control and data acquisition (SCADA) systems, equipment status indicators, protection-device actuation commands and fault alarms [6]. Such data directly support real-time control and rapid response to abnormal conditions.

By contrast, AP data underpin medium-term and long-term planning, decision analysis and risk assessment. These data are distinguished by large volume, multi-dimensional aggregation, batch processing and cross-domain correlation. Representative AP data encompass whole-life-cycle equipment records, power-grid topologies, regional load-forecast results and operation-and-maintenance risk-assessment reports [7]. Their value is extracted primarily through offline batch computation.

Fig. 1 compares TP and AP data across five dimensions: objective orientation, data characteristics, data volume, latency requirements and typical application scenarios. The pronounced divergence in these core attributes constitutes the principal obstacle to integration. Moreover, TP data are

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heterogeneous and distributed across multiple sources and granularities—ranging from equipment level to substation and regional levels—and appear in formats such as time-series streaming data and binary signalling data, necessitating cross-level data fusion.

Dimension	Transaction Processing (TP) Data	Analytical Processing (AP) Data
Objective Orientation	Derived from real-time operation processes of power systems	Serves planning and decision-making analysis of power systems
Data Characteristics	High concurrency, strict timeliness, small data packet size	Multi-dimensional aggregation, batch summarization, cross-domain correlation
Data Volume (Per Transaction)	Small	Massive
Latency Requirement	Millisecond-level	Minute-to-hour level
Typical Examples	Voltage and current measurements collected by SCADA systems, real-time monitoring signals, equipment fault alarms, etc.	Equipment full-lifecycle records, regional load forecasts, operation and maintenance (O&M) risk assessments, etc.

Fig. 1. A Comparative Analysis of the Specific Differences between TP and AP Data in the Smart Grid Domain.

In this work, we introduce **MGA-KGF**—a multi-granularity adaptive knowledge graph fusion framework for heterogeneous multi-source smart grid data. The framework adopts a layered architecture that enables end-to-end integration from data acquisition and knowledge modeling to graph fusion and intelligent applications, thereby offering a systematic solution for the efficient fusion of TP and AP data.

Our main contributions are summarized below:

- **We propose a spatio-temporally coupled, granularity-aware data-modeling method that resolves the multi-granularity entity-mapping problem in knowledge graphs.** Departing from traditional single-granularity modeling, the method constructs (i) fine-grained dynamic triples for the high-frequency real-time characteristics of TP data, (ii) coarse-grained scenario-oriented knowledge units for the batch statistical characteristics of AP data, and (iii) medium-grained knowledge as a collaborative bridge to achieve temporal alignment and spatial correlation between TP and AP data. Consequently, the fragmentation of knowledge between real-time transactional and offline analytical data is effectively mitigated.
- **We design a multi-level, event-driven, cross-domain knowledge-graph fusion and updating mechanism that enables bidirectional co-optimization of TP and AP data.** Specifically, we define three-level event triggers—device, station, and area—that jointly leverage the timeliness of TP data and the historical depth of AP data. A hierarchical alignment strategy is proposed: real-time data are aligned rapidly via the TransE algorithm, whereas historical batch data are matched in depth by a graph attention network (GAT). Moreover, temporal attention weights are introduced during entity embedding to capture the temporal decay of TP data, and statistical attention weights are employed to encode the confidence of AP data. These mechanisms optimize the entity-similarity computation model and effectively bridge the heterogeneous semantic gap.
- **We propose a load-aware, real-time knowledge-graph storage and indexing scheme that balances timeliness and storage efficiency.** Indexes are migrated dynamically according to data-access heat: hot data—e.g., recent TP real-time data—are encoded with gapped encoding to enhance query performance, whereas cold data—e.g., historical AP statistics—are compressed via succinct encoding to reduce storage overhead. A load-prediction module further adjusts entity granularity by forecasting future access patterns. During updates, fine-grained knowledge from TP data is refreshed in real time through stream processing, whereas coarse-grained knowledge from AP data is updated periodically via offline computation. Consequently, the knowledge graph achieves incremental synchronization of TP and AP data while maintaining low-latency access for TP data and high compression for AP data, thereby resolving the timeliness bottleneck inherent in conventional approaches.

II. Challenge

Although TP and AP data are complementary to the operation of smart grids, their in-depth integration still faces significant challenges. Traditional data fusion methods, such as rule-based matching methods and simple feature splicing algorithms, generally exhibit limitations including poor real-time performance, semantic fragmentation, and inadequate domain adaptability [8]. These limitations render them unsuitable for direct application in TP/AP data fusion scenarios, resulting in a lack of practical TP/AP data fusion algorithms and systematic frameworks in the smart grid field. Consequently, issues such as delayed fault diagnosis and insufficient accuracy in planning and decision-making arise, which further restrict the improvement of the overall operational efficiency of smart grids. Specifically, the current fusion of TP and AP data in smart grids mainly confronts the following five technical bottlenecks.

Insufficient real-time and concurrent processing capabilities. Equipment such as Supervisory Control and Data Acquisition (SCADA) systems generates massive volumes of multi-source heterogeneous TP data, with peak concurrent access reaching hundreds of thousands of times per second. However, the traditional data processing architecture struggles to meet the requirements of real-time data reception, parsing, and processing in this scenario, leading to data backlogs and processing delays [9].

Semantic barriers in multi-source heterogeneous data. TP data and AP data differ significantly in storage formats and data structures. TP data are mostly stored in real-time databases as time-series streams, while AP data are scattered across relational databases, document systems, and open Internet platforms [10]. Traditional fusion methods lack a unified semantic model, making it difficult to achieve semantic

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consistency between the two types of data and resulting in the formation of 'data silos'.

Inadequate representation of domain knowledge. Existing fusion methods fail to fully integrate the domain characteristics of smart grids to construct adaptive knowledge models, thus being unable to effectively capture the multi-dimensional features of grid data (e.g., equipment topology relationships and operational state correlations). Meanwhile, the relationships between grid entities are dynamically adjusted with operating conditions, exhibiting dynamic association complexity [11]. However, existing systems rely on manual experience to construct static association rules, which makes it difficult to update entity relationships in real time.

Poor quality of multi-granularity fusion. Standardized data processing procedures are lacking, and technical deficiencies exist in anomaly detection, format conversion, and semantic annotation during the data preprocessing stage [12]. These deficiencies lead to insufficient completeness, accuracy, and semantic reliability in the fusion of multi-granularity data, which in turn affects the accuracy of subsequent analysis results.

Mismatch in spatio-temporal granularity. TP data have a high time resolution (milliseconds or even nanoseconds) but limited spatial coverage. For instance, partial discharge monitoring sensors can capture nanosecond-level transient impulse signals, yet they are only deployed at key equipment nodes. In contrast, although AP data achieve wide-area spatial coverage, their time resolution is typically at the hourly level [13]. This asymmetry in spatio-temporal granularity makes it difficult to establish a precise causal chain of 'real-time monitoring data - historical statistics' during fault analysis, thereby reducing the accuracy of fault location and traceability.

III. Method

In accordance with the core principle of 'layered decoupling and dynamic adaptation', MGA-KGF abstracts complex, multi-source, heterogeneous smart-grid data into three logically independent yet jointly evolvable layers: the data-access layer, the knowledge-modeling layer, and the graph-fusion layer. Each layer is capable of self-description, self-governance, and self-evolution, and cross-layer coordination is realized through event-driven and load-aware mechanisms. Consequently, end-to-end knowledge fusion is achieved without disrupting existing business systems.

A. Data Access Layer: Unified Multi-source Access and Cross-modal Preprocessing

As the basic support module of the framework, the data access layer undertakes the tasks of unified reception, cleaning, and storage optimization for multi-source heterogeneous data. It realizes the initial integration of Transaction Processing (TP) and Analytical Processing (AP) data via differentiated processing strategies, laying a data foundation for subsequent knowledge modeling.

1) TP Data Access Module: Designed for real-time data sources (e.g., smart sensors, SCADA systems), this module adopts a high-concurrency stream processing architecture to stably receive over 100,000 access requests per second. Its workflow includes three key steps: first, cleaning raw data by eliminating abnormal values (e.g., extreme values exceeding equipment rated ranges) and missing values (caused by signal interruption) to ensure data authenticity; second, converting multi-format real-time data (e.g., binary signals, timing flow data) to JSON format to eliminate format differences; third, adding metadata annotations (e.g., device unique ID, acquisition timestamp) to each data entry, constructing a traceable real-time data chain to ensure accurate correlation of data sources and spatiotemporal attributes in subsequent knowledge modeling[14].

2) AP Data Access Module: This module mainly receives batch data from historical databases, including equipment full-life-cycle records, regional annual load statistics, and operation and maintenance (O&M) risk assessment reports. It implements two core operations: first, removing redundancy by comparing data identifiers and statistical dimensions (e.g., deleting duplicate regional load data in the same time period) to reduce storage redundancy; second, structuring cross-domain aggregated data (e.g., fused O&M records of multiple substations) by converting unstandardized text descriptions into a unified field format, providing a normalized data source for subsequent coarse-grained knowledge unit construction.

3) Anomaly Data Processing Module: To ensure data quality, this module adopts a 'statistical detection-intelligent identification' dual-mode strategy. For structured data (e.g., TP real-time measurement data, AP statistics), it combines the LSTM neural network and 3σ criterion: the LSTM model learns data temporal variation laws to predict normal data ranges, while the 3σ criterion screens outliers exceeding ±3 standard deviations of the mean, achieving complementarity between dynamic and static anomaly detection. For unstructured data (e.g., O&M logs, equipment fault images), natural language processing (NLP) technology extracts key entities (e.g., equipment ID, fault type, maintenance time) to convert unstructured text into structured information. For semi-structured data (e.g., XML-format grid topology model files), an adaptive parser identifies label hierarchies and data associations, converting them into an 'Entity-Relationship-Attribute' triad to provide a unified data format for knowledge modeling.

4) Storage Index Optimization Module: With 'dynamic adaptation-load prediction' as its core, this module balances data access efficiency and storage costs. It first implements dynamic index migration based on data query heat: high-frequency accessed recent TP real-time data (defined as hot data) uses Gapped encoding for index construction—reserving encoding gaps reduces index reconstruction overhead during data updates and improves real-time query response speed; long-term low-frequency accessed AP historical data (defined as cold data) uses Succinct encoding for compression storage, reducing storage occupancy while ensuring data integrity. Additionally, the module integrates a load prediction

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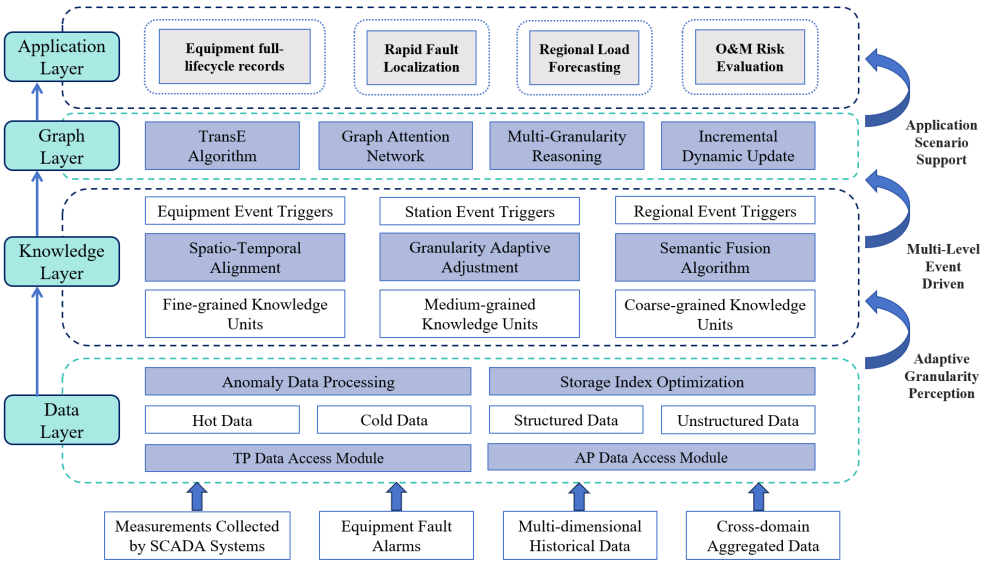


Fig. 2. MGA-KGF: A Multi-Granularity Adaptive Knowledge Graph Fusion Framework for Heterogeneous Multi-Source Smart Grid Data.

sub-module: by analyzing historical access records and grid business cycles, it predicts future data access patterns and dynamically adjusts entity granularity—splitting and storing high-frequency accessed fine-grained entities before access peaks to improve parallel query efficiency, and merging and storing low-frequency accessed coarse-grained entities during access valleys to save space.

B. Knowledge modeling Layer: Multi-Granularity Knowledge Construction and Spatio-Temporal Semantic Collaboration

As the core of the MGA-KGF framework, the knowledge modeling layer constructs fine-grained, medium-grained, and coarse-grained knowledge units. It integrates spatio-temporal alignment and semantic fusion technologies to convert TP and AP data into knowledge, thereby breaking semantic barriers and granularity differences between the two data types.

1) Definition and Generation of Multi-Granularity Knowledge Units: Based on the characteristic differences of smart grid data, the knowledge modeling layer defines criteria and generation rules for fine-grained, medium-grained, and coarse-grained knowledge units, forming a knowledge system covering equipment, station, and regional levels. The three types of units and their generation mechanisms are as follows.

Fine-grained knowledge units: Dominated by TP data, they focus on real-time state description at the equipment level, with the core structure of the 'Entity-Relationship-Attribute-Spatio-Temporal Label' triad. During generation, real-time measurements and control commands are extracted from the TP data access layer, associated with equipment IDs and collection timestamps to form accurate real-time knowledge records, which directly support equipment real-time monitoring and abnormal response.

Medium-grained knowledge units: Constructed from both TP and AP data, they focus on station-level business scenario association, with the core structure of 'Business Scenario-Entity Set-Association Rules'. During generation, fine-grained TP data of multiple substation equipment and in-station AP historical data are integrated; scenario features are identified via semantic matching, and inter-entity association logic is refined. These units enable linkage analysis of equipment status and business scenarios, supporting initial fault localization.

Coarse-grained knowledge units: Dominated by AP data, they focus on regional-level decision support, with the core structure of 'Region-Statistical Indicators-Decision Rules'. During generation, regional batch data from the AP data access layer are statistically analyzed; combined with grid operation standards and historical decision cases, regional-level knowledge is constructed to support regional power grid planning and macro decision-making.

The three types of knowledge units are not independent. Fine-coarse granularity synergy is achieved through medium-grained units: for example, real-time transformer voltage fluctuations (fine-grained) are associated with regional load prediction data (coarse-grained) to establish the 'Equipment State-Regional Load' medium-grained association rule. This rule pre-warns equipment overload risks during regional load growth, improving fault prediction accuracy.

2) Spatio-Temporal Alignment: To address spatio-temporal granularity differences between TP and AP data, the knowledge modeling layer adopts the Dynamic Time Warping (DTW) algorithm and spatial topology mapping technology to realize spatio-temporal synergy of the two data types.

Temporal dimension: The DTW algorithm calculates time series similarity between TP and AP data to automatically adjust the time axis, eliminating temporal resolution deviations between the two data types and ensuring accurate matching between real-time data temporal features and statistical data

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time nodes.

Spatial dimension: A spatial mapping model is constructed based on power grid topology, associating equipment-level spatial information (from TP data) with regional spatial information (from AP data). This forms an 'Equipment-Substation-Region' spatial hierarchy chain, ensuring spatial consistency of multi-granularity data at the knowledge level and laying the foundation for subsequent cross-granularity fusion.

3) Semantic Fusion: The semantic fusion module achieves semantic unification of transactional data (TP) and analytical data (AP) by constructing a smart grid domain ontology layer. Defined per the Web Ontology Language (OWL) specification, this layer covers core grid entities (equipment, substation, region), relationships (real-time monitoring, statistical analysis, control association), attributes (voltage, line loss rate, fault type), and constraint rules (e.g., equipment rated voltage range, line loss rate qualification standard), forming a unified domain semantic standard. During semantic mapping, the module associates TP data terms (e.g., 'real-time measurement value', 'control command') with the ontology layer's 'real-time monitoring attribute' and 'control relationship', and AP data terms (e.g., 'annual load', 'operation and maintenance report') with the ontology layer's 'statistical attributes' and 'decision support'. To resolve semantic ambiguity, domain rule-based semantic disambiguation is implemented to ensure semantic consistency between TP and AP data [15].

4) Granularity Adaptive Adjustment: Based on grid operation status and business requirements, this module dynamically optimizes knowledge unit granularity. It triggers the granularity adjustment mechanism by monitoring three factors in real time: data access frequency, business scenario complexity (e.g., fine-grained data for fault diagnosis, coarse-grained data for regional planning), and data volume changes. Specific adjustment logic includes: (1) When equipment faults occur in a region, the regional knowledge unit is automatically split from coarse-grained (regional) to medium-grained (substation-level) and fine-grained (equipment-level) to provide accurate fault-related data; (2) After fault resolution, if data access frequency decreases, fine-grained knowledge units are merged into medium-grained ones to reduce storage and computation overheads; (3) Combined with load forecasting results, regional load-related knowledge units are split into fine-grained ones before electricity consumption peaks to support high-frequency load monitoring, and merged into coarse-grained ones during low-consumption periods to optimize resource allocation.

5) Three-Level Event Triggering Mechanism: The three-level event triggering mechanism adopts an 'equipment-station-region' hierarchy to realize dynamic knowledge unit updates and business linkage, with the following workflow: (1) Equipment-level trigger: Monitors real-time equipment status changes in TP data; upon detecting anomalies, it immediately updates fine-grained knowledge units and sends warning signals to the station-level trigger; (2) Station-level trigger: Integrates abnormal signals from multiple in-station devices, judges whether a station-level fault scenario is formed; if a station-level event is triggered, it updates medium-grained knowledge units and reports to the regional trigger; (3) Regional-level trigger: Analyzes event information from multiple substations, judges whether a regional-level event is formed; if a regional-level event is triggered, it updates coarse-grained knowledge units and pushes them to the decision support module. The three-level triggers collaborate to achieve rapid responses using real-time TP data, and verify event authenticity by incorporating AP data's historical depth—this avoids false triggers and ensures the accuracy and timeliness of knowledge updates.

C. Graph Fusion Layer: Cross-domain Fusion and Dynamic Reasoning

As the application support module of the MGA-KGF framework, the Graph Fusion Layer achieves in-depth knowledge integration and efficient application via cross-domain data fusion, multi-granularity hierarchical reasoning, and incremental dynamic update technologies, providing decision support for smart grid business scenarios.

1) Cross-domain Data Fusion: This module adopts a 'TransE-GAT' dual-mode fusion strategy to realize entity association and semantic unification of TP (real-time) and AP (historical) data. For TP real-time data: The TransE algorithm is used to construct an entity embedding model. Fine-grained entities (e.g., devices, real-time attributes) are mapped to a low-dimensional vector space, and real-time entity alignment is accelerated by calculating vector distances to meet low-latency requirements of real-time fusion. For AP historical data: A Graph Attention Network (GAT) is used to build a deep fusion model. Differentiated weights are assigned to different AP entities via the attention mechanism to learn deep inter-entity associations and achieve accurate coarse-grained entity matching.

In cross-type fusion, the GAT model maps fine-grained TP entities and coarse-grained AP entities to a unified vector space; entity correlation is quantified using cosine similarity. Temporal and statistical attention weights are introduced in entity embedding. Temporal attention weights decay dynamically with TP data timeliness to capture TP's temporal characteristics. Statistical attention weights are assigned based on AP data sample size and confidence to encode AP's confidence level.

This mechanism optimizes the entity similarity calculation model, bridges heterogeneous semantic gaps between TP and AP data, and enables deep cross-domain knowledge integration.

2) Multi-Granularity Hierarchical Reasoning: This module adopts differentiated reasoning strategies based on the distinct application requirements of TP and AP data to ensure accurate knowledge value output. For TP-supported real-time services: A 'k-hop neighbor query' strategy is adopted. Combined with G-Tree global indexing and Locality-Sensitive Hashing (LSH) technology, associated entities of faulty equipment are quickly located, and their real-time states are acquired to achieve second-level fault-associated entity positioning and support

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real-time fault diagnosis. For AP-supported offline services: Graph diffusion algorithms (e.g., random walk with restart) are used for in-depth reasoning. Combined with decision rules in coarse-grained knowledge units, operation and maintenance proposals and planning schemes are generated to support long-term decision-making for regional power grids.

3) Incremental Dynamic Update: This module realizes real-time synchronization and efficient update of TP and AP knowledge graphs (KGs). For TP data: It monitors incremental changes in the TP data access layer in real time, converts these changes into KG incremental update commands, and writes them to the KG in real time to maintain up-to-date fine-grained knowledge units. For AP data: An offline batch update strategy is adopted. Incremental statistical data are periodically extracted from the AP data access layer; coarse-grained knowledge units are updated via batch processing tools, and consistency with fine-grained knowledge units is synchronously verified.

An update conflict resolution mechanism is introduced: When TP and AP update contents conflict, the 'real-time priority-historical verification' principle is applied [16]. TP real-time data is prioritized for KG update, while AP data statistical processes are traced to identify conflict causes and correct AP data—ensuring KG consistency and accuracy. Incremental dynamic updates enable synchronous iteration of TP and AP knowledge, avoiding the timeliness bottleneck of traditional static KGs.

IV. Application

Through multi-granular knowledge fusion and dynamic reasoning, the MGA-KGF framework can be deeply applied in the core business scenarios of smart grid, such as equipment management, fault handling, load scheduling, risk prevention and control, which can effectively improve the efficiency of grid operation and scientific decision-making.

A. Equipment Lifecycle Digital Ledger

With the help of multi-granular knowledge graph, MGA-KGF integrates heterogeneous data scattered in more than 10 business systems, such as design parameters, factory tests, operation and monitoring, maintenance records, and decommissioning assessment, into 'a single digital map for each piece of equipment'. Designers can view macro technical specifications, operation and maintenance personnel can track real-time working conditions, and asset management personnel can assess the remaining life of the equipment. All perspectives share the same set of semantic baselines, avoiding information disconnection and duplicate entries.

B. Rapid Fault Location and Traceability

When a protective device operates, the event bus triggers subgraph matching at the station level in milliseconds, and the system quickly converges on the set of suspicious devices based on real-time measurements and historical defect records; subsequently, the graph inference engine combines regional statistical laws to locate the fault propagation path and generate a complete evidence chain of 'fault-anomaly-cause' [17].

C. Regional Load Forecasting and Resilience Assessment

By correlating real-time measurements at the device level with multi-dimensional AP data at the regional level, such as economic, meteorological, and holiday data, MGA-KGF constructs a load evolution map with temporal and spatial consistency. The prediction model captures sudden change signals at the fine-grained end and absorbs long-term trends at the coarse-grained end, thus significantly improving the prediction accuracy under extreme weather, emergencies and other scenarios. Further, the system can simulate 'N-1' and even 'N-k' fault scenarios to quantify the resilience margin of the regional grid and provide a quantitative basis for the adjustment of operation mode[18].

D. Operation and Maintenance Risk Assessment and Overhaul Strategy Optimization

Based on multi-dimensional indicators such as equipment health, historical failure rate, load importance and maintenance resource constraints, the system automatically generates risk level scores in the knowledge graph and gives maintenance priority ranking through graphical reasoning. Maintenance planners can visually view the risk-cost curves under different strategies, thus achieving optimal allocation of maintenance resources while ensuring power supply reliability [19].

V. Conclusion

In this paper, we present MGA-KGF, a multi-granularity, layered framework for fusing heterogeneous Transaction Processing (TP) and Analytical Processing (AP) data in smart grids. At the data level, MGA-KGF unifies multi-source access and quality assurance; at the knowledge level, it constructs fine-grained, medium-grained, and coarse-grained knowledge units with explicit spatio-temporal collaboration; and at the graph level, it realises bidirectional TP/AP fusion through an event-driven mechanism and a dual-mode TransE–GAT pipeline enhanced by temporal and statistical attention. A load-aware storage and indexing scheme ensures low-latency queries on hot, real-time knowledge while maintaining compact storage for cold, historical knowledge.

Across representative grid tasks—including equipment lifecycle management, rapid fault localisation, regional load forecasting, and O&M risk assessment—MGA-KGF consistently improves timeliness, semantic consistency, and cross-granularity reasoning quality, demonstrating both engineering practicality and methodological generality. Its design decouples data access, knowledge modelling, and graph fusion, making it extensible to new modalities and business scenarios.

Future work will focus on: (i) leveraging large language models for ontology construction, semantic disambiguation, and explainable reasoning; (ii) extending to inverter-dominated, renewable-heavy grids with fast-changing, non-stationary conditions; and (iii) building open benchmarks and pipelines for online evaluation and continual learning to enhance robustness and replicability.

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VI. ACKNOWLEDGEMENT

This work was supported by the Science and Technology Project of China Southern Power Grid Co., Ltd.(ZBKJXM20220010).

REFERENCES

[1] Z. Shao, M. Wang, et al. "Real-time dynamic voltage loop scheduling for multi-core embedded systems". In: *IEEE Trans. on Circuits and Systems II* 54.5 (2007), pp. 445–449.

[2] M. Qiu, W. Dai, and A. Vasilakos. "Loop parallelism maximization for multimedia data processing in mobile vehicular clouds". In: *IEEE T. on Cloud Computing* 7.1 (2016), pp. 250–258.

[3] Y. Gao, S. Iqbal, et al. "Performance and power analysis of high-density multi-GPGPU architectures: A preliminary case study". In: *IEEE 17th HPCC*. 2015.

[4] J. Li, M. Qiu, et al. "Thermal-aware task scheduling in 3D chip multiprocessor with real-time constrained workloads". In: *ACM Trans. on Embedded Comp. Sys. (TECS)* 12.2 (2013), pp. 1–22.

[5] Dabeeruddin Syed et al. "Smart Grid Big Data Analytics: Survey of Technologies, Techniques, and Applications". In: *IEEE Access* 9 (2021), pp. 59564–59585.

[6] Hamed Taherdoost. "A systematic review of big data innovations in smart grids". In: *Results in Engineering* 22 (2024), p. 102132.

[7] Amr A. Munshi and Yasser Abdel-Rady I. Mohamed. "Data Lake Lambda Architecture for Smart Grids Big Data Analytics". In: *IEEE Access* 6 (2018), pp. 40463–40471.

[8] M Talaat et al. "Hybrid-cloud-based data processing for power system monitoring in smart grids". In: *Sustainable Cities and Society* 55 (2020), p. 102049.

[9] Kun Wang et al. "Wireless big data computing in smart grid". In: *IEEE Wireless Communications* 24.2 (2017), pp. 58–64.

[10] Ameema Zainab et al. "Big data management in smart grids: Technologies and challenges". In: *IEEE Access* 9 (2021), pp. 73046–73059.

[11] Olufemi A Omitaomu and Haoran Niu. "Artificial intelligence techniques in smart grid: A survey". In: *Smart Cities* 4.2 (2021), pp. 548–568.

[12] Song Tan et al. "Survey of security advances in smart grid: A data driven approach". In: *IEEE Communications Surveys & Tutorials* 19.1 (2016), pp. 397–422.

[13] Chatura Seneviratne, Patikiri Arachchige Don Shehan Nilmantha Wijesekara, and Henry Leung. "Performance analysis of distributed estimation for data fusion using a statistical approach in smart grid noisy wireless sensor networks". In: *Sensors* 20.2 (2020), p. 567.

[14] Yi Wu et al. "Efficient multi-source self-attention data fusion for fdia detection in smart grid". In: *Symmetry* 15.5 (2023), p. 1019.

[15] Dina Fawzy, Sherin M Moussa, and Nagwa L Badr. "An IoT-based resource utilization framework using data fusion for smart environments". In: *Internet of Things* 21 (2023), p. 100645.

[16] Rustem Dautov, Salvatore Distefano, and Rajkumaar Buyya. "Hierarchical data fusion for smart healthcare". In: *Journal of Big Data* 6.1 (2019), pp. 1–23.

[17] Andrei BB Torres et al. "Multilevel data fusion for the internet of things in smart agriculture". In: *Computers and Electronics in Agriculture* 171 (2020), p. 105309.

[18] Chr Lamnatou, D Chemisana, and C Cristofari. "Smart grids and smart technologies in relation to photovoltaics, storage systems, buildings and the environment". In: *Renewable Energy* 185 (2022), pp. 1376–1391.

[19] Nikoleta Andreadou, Miguel Olariaga Guardiola, and Gianluca Fulli. "Telecommunication technologies for smart grid projects with focus on smart metering applications". In: *Energies* 9.5 (2016), p. 375.

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